F7AB - Actuarial and Financial Mathematics
Tutorial 6 – Yields and Measuring fund performance

Question 1 You pay a price of £1 000 for an investment that will repay you £120 at the end of each of the next 10 years. Calculate the yield on this investment.
Question 2 You invest £10 000 in a project and receive back £700.87 per annum payable half-yearly in arrears for the next 25 years. Calculate the yield on this investment.
Question 3 You invest £100 000 in a project which returns £20 000 at the end of each of the first, second and third years, and then £40 000 at the end of each of the fourth, fifth and sixth years. Calculate the yield on this project.
Question 4 You invest in a mining project which requires you to pay £1 000 000 at the start of each of the calendar years 2018, 2019 and 2020 to develop the mine. It is expected to be in active production for twenty years, returning a profit of £250,000 at the end of each calendar year 2024 to 2043 inclusive. Calculate the yield on this project.
Question 5 Assume the set-up of Question 4 of Tutorial 5.

1. In each case considered in Q4.1, write down the equation of value at the end of the loan period and then verify that the yield on the transaction is still 6.5% per annum.

2. In the case considered in Q4.3, write down the equation of value at the end of the loan period and then calculate the yield on the transaction.

Question 6 An investor is considering investing in two possible projects.

- Project A is a smartphone factory, which will cost £200 million immediately to build, and at the end of each of the first, second, third, fourth and fifth years will make a profit of £60 million. At the end of the fifth year the current generation of mobile telephones will be obsolete, and the factory will be sold for £20 million.

- Project B is identical to Project A until the end of the fifth year. At that time the factory will make a profit of £60 million as before, but instead of being sold the factory will be re-tooled at a cost of £180 million to produce the next generation of smartphones. It will then make a profit of £60 million at the end of the sixth, seventh, eighth, ninth and tenth years, and at the end of the tenth year it will be sold for £20 million.

1. Calculate the yield on Project A.

2. Calculate a yield on Project B.

3. State sufficient conditions, involving the signs of the net cash-flows under a financial transaction, under which there exists a unique yield $i^\star \geq -1$ on that transaction.

4. Verify that Project A satisfies the conditions above, but that Project B does not.

5. Prove that the yield on Project B found in part (2) above is unique, despite part (4) above.

Question 7 You have £1 000 000 to invest, and you have a choice of two investments.

- Investment A will return £130 000 at the end of each of the next 10 years, and in addition will return your original investment of £1 000 000 at the end of the 10 years.
- Investment B will return a series of payments, annually in arrears, starting at £160 000 at the end of the first year, then increasing by 5% per annum compound until the final payment is made at the end of the 10th year.

Using the highest yield as the decision-making criterion, determine the investment that you would choose.

Question 8 Suppose you have two series of cashflows:

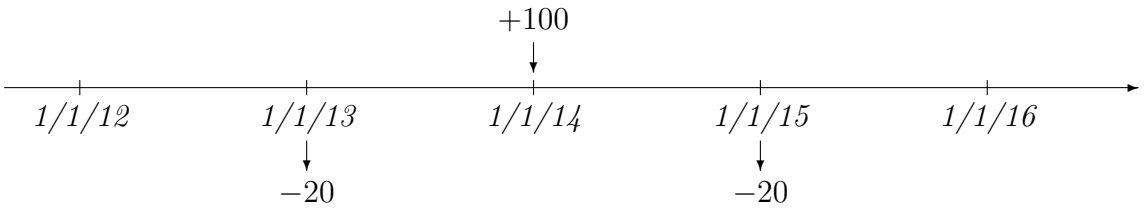
- cashflows of $a_1, a_2, \ldots, a_n$ are payable at times $t_1, t_2, \ldots, t_n$, and
- cashflows of $b_1, b_2, \ldots, b_m$ are payable at times $s_1, s_2, \ldots, s_m$,

for strictly positive integers n and m. Suppose that the yields on both sets of cashflows exist and are equal to the same value $i^\star$.

Show that if you combine the two sets of cashflows into a single transaction, then $i^\star$ is also a solution of the equation of value of the new, combined cashflows.

However, setting $n = m = 2$, find an example to show that it need not be the unique such solution ≥ -1.

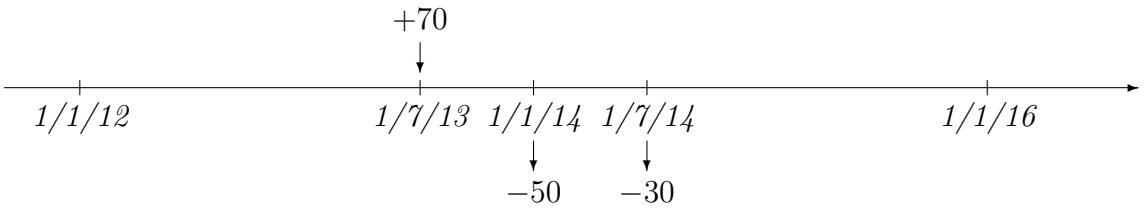
Question 9 A fund, which has value £100 on 1 January 2012, has the following cashflows in £:



The value of the assets in the fund on 1 January 2016 is £170.

1. Calculate the Money-Weighted Rate of Return (MWRR) during the calendar years 2012–2015.
2. If the fund value on 1 January 2014 was £100 just before the cash inflow on 1 Jan 2014 was received, calculate the Linked Internal Rate of Return (LIRR) during the calendar years 2012–2015, based on the two sub-periods calendar years 2012–2013 and 2014–2015.
3. If, in addition, the fund value was £110 on 1 January 2013 and £180 on 1 January 2015 (just before the cashflows on 1 January 2013 and 2015), calculate the Time-Weighted Rate of Return (TWRR) during the calendar years 2012–2015.

Question 10 A fund, which has value £100 on 1 January 2012, has the following cashflows in £:



The value of the assets in the fund on 1 January 2016 is £140.

1. Calculate the Money-Weighted Rate of Return (MWRR) during the calendar years 2012–2015.

2. If the fund value on 1 January 2013 was £105, and on 1 January 2015 was £130, calculate the Linked Internal Rate of Return (LIRR) during the calendar years 2012–2015, based on the three sub-periods 2012, 2013–2014 and 2015.

3. Suppose, in addition to the information in part 2, that you knew the following fund values:

Date	1 July 2013	1 January 2014	1 July 2014
Fund Value	£115	£195	£155

   Calculate the Time-Weighted Rate of Return (TWRR) during the calendar years 2012–2015.

Question 11 [CT1, Apr 2010, Q10] A pension fund's assets were invested with two fund managers. On 1 January 2007, Manager A was given £120 000 and Manager B was given £100 000. A further £10 000 was invested with each manager on 1 January 2008 and again on 1 January 2009.

The values of the funds were:

Date	31 Dec 2007	31 Dec 2008	31 Dec 2009
Manager A	£130 000	£135 000	£180 000
Manager B	£140 000	£145 000	£150 000

1. Calculate the time-weighted rates of return earned by Manager A and Manager B over the period 1 January 2007 to 31 December 2009.

2. Show that the money-weighted rate of return earned by Manager A over the period 1 January 2007 to 31 December 2009 is approximately 9.4% per annum.

3. Explain, without performing further calculations, whether the money-weighted rate of return earned by Manager B over the period 1 January 2007 to 31 December 2009 was higher than, lower than or equal to that earned by Manager A.

4. Discuss the relative performance of the two fund managers.

Question 12 [CT1, Sep 2011, Q4] A pension fund makes the following investments in £m:

Date	1 Jan 2009	1 Jul 2009	1 Jan 2010
Investment amounts (£m)	1.5	6.0	4.0

The rates of return earned on money invested in the fund were as follows:

Time period	Rate of return (i.e. not annualised)
1 Jan 2009 to 30 Jun 2009	1%
1 Jul 2009 to 31 Dec 2009	2%
1 Jan 2010 to 31 Dec 2010	5%

Assume that 1 January to 30 June and 1 July to 31 December are precise half-year periods.

1. Calculate the time-weighted rate of return per annum effective over the two years from 1 January 2009 to 31 December 2010.

2. Calculate the money-weighted rate of return per annum effective over the two years from 1 January 2009 to 31 December 2010.

Answers: Q1. 0.0345; Q2. 0.0498; Q3. 0.1679; Q4 0.0344; Q5.2. 0.0668; Q6.1 0.1722; Q6.2 0.1722; Q7. B; Q9.1. 1.878%; Q9.2. 3.706%; Q9.3. 3.975%; Q10.1. 9.973%; Q10.2. 9.652%; Q10.3. 9.758%; Q11.1 A: 9.050% p.a. B: 9.409% p.a.; Q11.3. Lower; Q12.1. 4.005% p.a.; Q12.2 4.63% p.a

F7AB - Actuarial and Financial Mathematics
Solutions to Tutorial 6 – Yields and Measuring fund performance

Solution 1 The equation of value is

$$f(i) := -1\,000\,(1+i)^{10} + 120\,s_{\overline{10}|}^{i} = 0.$$

As we cannot solve the equation of value exactly, we use trial-and-error. To get a starting value for the trial-and-error method, we do a first order approximation of the equation of value:

$$f(i) \approx -1\,000\,(1+10i) + 120\left(\frac{1+10i-1}{i}\right) = 0 \quad \Rightarrow \quad i \approx \frac{200}{10\,000} = 0.02.$$

We start by evaluating $f(0.02)$, $f(0.03)$, and so on until we find two values where the sign of f changes. Using this method, we calculate $f(0.03) = +31.75$ and $f(0.04) = -39.51$. We seek $i^\star$ such that $f(i^\star) = 0$ and assume that f is linear between $i = 0.03$ and $i = 0.04$, i.e.

$$\frac{i^\star - 0.03}{0.04 - 0.03} = \frac{f(i^\star) - f(0.03)}{f(0.04) - f(0.03)} \quad \Rightarrow \quad i^\star \approx 0.03 + 0.01\,\frac{31.75}{31.75 + 39.51} = 0.0345.$$

Solution 2 Using a time unit of a half-year, the equation of value is

$$f(i) := -10\,000\,(1+i)^{50} + \frac{700.87}{2}\,s_{\overline{50}|}^{i} = 0.$$

As we cannot solve the equation of value exactly, we use trial-and-error. To get a starting value for the trial-and-error method, we do a first order approximation of the equation of value:

$$f(i) \approx -10\,000\,(1+50i) + \frac{700.87}{2}\left(\frac{1+50i-1}{i}\right) = 0 \quad \Rightarrow \quad i \approx \frac{7521.75}{500\,000} = 0.015.$$

We start by evaluating $f(0.015)$, $f(0.02)$, and so on until we find two values where the sign of f changes. We find $f(0.02) = +2723.702$ and $f(0.025) = -209.159$. We seek $i^\star$ such that $f(i^\star) = 0$ and assume that f is linear between $i = 0.02$ and $i = 0.025$, i.e.

$$\frac{i^\star - 0.02}{0.025 - 0.02} = \frac{f(i^\star) - f(0.02)}{f(0.025) - f(0.02)} \quad \Rightarrow \quad i^\star \approx 0.02 + 0.005\,\frac{2723.702}{2723.702 + 209.159} = 0.0246.$$

Hence in time units of 1 year, the yield is approximately 0.0498. (The exact solution is 0.05.)

Solution 3 In thousands of pounds, the equation of value is

$$f(i) := -100\,(1+i)^{6} + 20\,s_{\overline{6}|}^{i} + 20\,s_{\overline{3}|}^{i} = 0.$$

As we cannot solve the equation of value exactly, we use trial-and-error. To get a starting value for the trial-and-error method, we do a first order approximation of the equation of value:

$$f(i) \approx -100\,(1+6i) + 20\left(\frac{1+6i-1}{i}\right) + 20\left(\frac{1+3i-1}{i}\right) = 0 \quad \Rightarrow \quad i \approx \frac{80}{600} = 0.013.$$

We start by evaluating $f(0.013)$, $f(0.14)$, and so on until we find two values where the sign of f changes. We find $f(0.16) = +6.022$ and $f(0.17) = -1.601$. We seek $i^\star$ such that $f(i^\star) = 0$ and assume that f is linear between $i = 0.016$ and $i = 0.017$, i.e.

$$\frac{i^\star - 0.16}{0.17 - 0.16} = \frac{f(i^\star) - f(0.16)}{f(0.17) - f(0.16)} \quad \Rightarrow \quad i^\star \approx 0.16 + 0.01\,\frac{6.022}{6.022 + 1.601} = 0.1679.$$

Solution 4 Be careful to distinguish between 'start of year' and 'end of year'! In thousands of pounds, the equation of value is

$$f(i) = -1\,000\,\ddot{s}^{i}_{\overline{3}|}\,(1+i)^{23} + 250\,s^{i}_{\overline{20}|} = 0.$$

As we cannot solve the equation of value exactly, we use trial-and-error. To get a starting value for the trial-and-error method, we do a first order approximation of the equation of value in which we also approximate $d \approx i$:

$$f(i) \approx -1\,000\,\frac{1+3i-1}{i}\,(1+23i) + 250\left(\frac{1+20i-1}{i}\right) = 0 \quad \Rightarrow \quad i \approx \frac{2\,000}{69\,000} = 0.029.$$

We start by evaluating $f(0.03)$, $f(0.04)$, and so on until we find two values where the sign of f changes. We find $f(0.03) = +434.43$ and $f(0.04) = -557.09$. We seek $i^\star$ such that $f(i^\star) = 0$ and assume that f is linear between $i = 0.03$ and $i = 0.04$, i.e.

$$\frac{i^\star - 0.03}{0.04 - 0.03} = \frac{f(i^\star) - f(0.03)}{f(0.04) - f(0.03)} \quad \Rightarrow \quad i^\star \approx 0.03 + 0.01\,\frac{434.43}{434.43 + 557.09} = 0.0344.$$

Solution 5

1. From Question 4 of Tutorial 5, the original repayment is $X^{(0)} = £3\,354.91$.

   Case (a): From Question 4 of Tutorial 5, the new repayment is $X^{(1)} = £4\,785.99$. Hence the equation of value is:

   $$f(i) := 35\,000\,(1+i)^{18} - 3\,354.91\,s^{i}_{\overline{10}|}\,(1+i)^{8} - 4\,785.99\,s^{i}_{\overline{6}|} = 0.$$

   It is verifiable by direct substitution in to the equation of value that $i^\star = 0.065$ is a solution of this.

   Case (b): From Question 4 of Tutorial 5, the new repayment is $X^{(2)} = £2\,656.67$. Hence the equation of value is:

   $$f(i) := 35\,000\,(1+i)^{21} - 3\,354.91\,s^{i}_{\overline{10}|}\,(1+i)^{11} - 2\,656.67\,s^{i}_{\overline{11}|} = 0.$$

   It is verifiable by direct substitution in to the equation of value that $i^\star = 0.065$ is a solution of this.

2. From Question 4 of Tutorial 5, the new repayment is $X^{(4)} = £5\,013.32$.

   Hence the equation of value is:

   $$f(i) := 35\,000\,(1+i)^{18} - 3\,354.91\,s^{i}_{\overline{10}|}\,(1+i)^{8} - 5\,013.32\,s^{i}_{\overline{6}|} = 0.$$

   As we cannot solve the equation of value exactly, we use trial-and-error to find an approximate solution. We know that the yield must be greater than 6.5 per cent p.a. because the lender is receiving larger repayments.

   As $f(0.065) = -1\,605.74$ and $f(0.07) = +2\,792.91$, and we seek $i^\star$ such that $f(i^\star) = 0$,

   $$\frac{i^\star - 0.065}{0.07 - 0.065} = \frac{f(i^\star) - f(0.065)}{f(0.07) - f(0.065)} \quad \Rightarrow \quad i^\star \approx 0.065 + 0.005\,\frac{1\,605.74}{1\,605.74 + 2\,792.91} = 0.0668.$$

Solution 6

1. The cashflows under Project A (in £million) are:

Time	0	1	2	3	4	5
Cashflow	-200	$+60$	$+60$	$+60$	$+60$	$+80$

The equation of value is:

$$f_A(i) := -200\,(1+i)^5 + 60\,s_{\overline{5}|}^{i} + 20 = 0.$$

As we cannot solve the equation of value exactly, we use trial-and-error. As $f_A(0.17) = +2.374$ and $f_A(0.18) = -8.299$, and we seek $i^\star$ such that $f_A(i^\star) = 0$,

$$\frac{i^\star - 0.17}{0.18 - 0.17} = \frac{f_A(i^\star) - f_A(0.17)}{f_A(0.18) - f_A(0.17)} \quad \Rightarrow \quad i^\star \approx 0.17 + 0.01\,\frac{2.374}{2.374 + 8.299} = 0.1722.$$

2. The cashflows under Project B (in £million) are:

Time	0	1	2	3	4	5	6	7	8	9	10
Cashflow	-200	$+60$	$+60$	$+60$	$+60$	$+60-180$ $=+80-200$	$+60$	$+60$	$+60$	$+60$	$+80$

We can solve directly for the yield on Project B. However, it is quicker to note that Project B can be decomposed into two series of cashflows, in both of which the amounts of cashflow are the same as for Project A.

Hence the equation of value for Project B can be written as:

$$f_B(i) := f_A(i)\,(1+i)^5 + f_A(i) = 0 \qquad \Rightarrow \qquad f_B(i) := f_A(i)\left((1+i)^5 + 1\right) = 0.$$

To solve $f_B(i) = 0$, we solve either $f_A(i) = 0$ or $(1+i)^5 + 1 = 0$. From the last part of the question, $f_A(0.1722) \approx 0$. So a yield on Project B is 0.1722.

3. From a result in the lecture notes, a series of cashflows has a well-defined yield (a unique real solution ≥ -1 of the equation of value) if all the positive cashflows precede all the negative cashflows, or *vice versa*.

4. It is clear that Project A satisfies the conditions in part (3) above, but that the net cashflow of -120 at $t = 5$ under Project B means that Project B does not.

5. From part (2) above, we can write:

$$f_B(i) = f_A(i)\,\left((1+i)^5 + 1\right) = 0.$$

As $i^\star$ is a well-defined yield for Project A, any real solution ≥ -1 other than $i^\star$ must be a solution of $(1+i)^5 + 1 = 0$. But this equation has no real roots ≥ -1. Hence $i^\star$ is indeed the yield on Project B.

Solution 7 The yield on Investment A is 13% p.a. effective.

In thousands of pounds, the equation of value for Investment B (in £000s) is

$$\begin{aligned}
f(i) &:= -1\,000\,(1+i)^{10} + 160\sum_{k=1}^{10} 1.05^{k-1}\,(1+i)^{10-k} = 0 \\
\Rightarrow 0 &= -1\,000\,(1+i)^{10} + 160 \times 1.05^9 \times \sum_{k=1}^{10} 1.05^{k-10}\,(1+i)^{10-k} \\
&= -1\,000\,(1+i)^{10} + 160 \times 1.05^9 \times \sum_{k=1}^{10} \left(\frac{1+i}{1.05}\right)^{10-k} \\
&= -1\,000\,(1+i)^{10} + 160 \times 1.05^9 \times s_{\overline{10}|}^{i'},
\end{aligned}$$

where $i' := (1+i)/(1.05) - 1$.

As we cannot solve the equation of value exactly, we use trial-and-error. As $f(0.13) = +136.778$ and $f(0.14) = -12.418$, and we seek $i^\star$ such that $f(i^\star) = 0$,

$$\frac{i^\star - 0.13}{0.14 - 0.13} = \frac{f(i^\star) - f(0.13)}{f(0.14) - f(0.13)} \quad \Rightarrow \quad i^\star \approx 0.13 + 0.01\,\frac{136.778}{12.418 + 136.778} = 0.1392.$$

Hence you would choose Investment B as it results in the highest yield.

Solution 8 Let $f_1(i) = 0$ and $f_2(i) = 0$ be the two equations of value associated with the first and second series of cashflows, respectively, both at time 0 rather than at time t_n and s_m, respectively. That is:

$$\begin{aligned} f_1(i) &:= a_1\,v^{t_1} + a_2\,v^{t_2} + \cdots + a_n\,v^{t_n} = \sum_{k=1}^{n} a_r\,v^{t_k} = 0, \\ f_2(i) &:= b_1\,v^{s_1} + b_2\,v^{s_2} + \cdots + b_m\,v^{s_m} = \sum_{\ell=1}^{m} b_r\,v^{s_\ell} = 0. \end{aligned}$$

and we are told that $f_1(i^\star) = 0$ and $f_2(i^\star) = 0$.

If we create a new set of cashflows simply by combining the two sets, the new equation of value at time 0 is

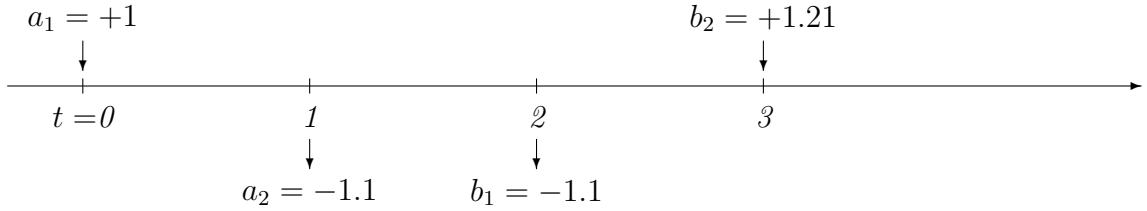
$$f(i) := f_1(i) + f_2(i) = 0.$$

Then

$$f(i^\star) = f_1(i^\star) + f_2(i^\star) = 0 + 0 = 0,$$

i.e. $i^\star$ satisfies the equation of value for the combined set.

Now we look for an example to show that $i^\star$ need not be the unique such solution ≥ -1. Consider the following sets of cashflows:



The two equations of value are:

$$f_1(i) := (1+i) - 1.1 = 0 \qquad \text{and} \qquad f_2(i) := -1.1\,(1+i) + 1.21 = 0.$$

It is easily checked that $i^\star = 0.1$ is the unique real solution ≥ -1 of each of these. Notice that $f_2(i) = -1.1\,f_1(i)$, so when we write down the equation of value of the combined series of cashflows,

$$f(i) = f_1(i) + f_2(i) = f_1(i)\left(1 - 1.1v^2\right) = 0.$$

Then for $f(i) = 0$, either $f_1(i) = 0$ or $1 - 1.1v^2 = 0$. Solving the latter equation gives a second real solution $j = 1.1^{1/2} - 1 \approx 0.0488088 \geq -1$. Hence, while there is a solution to the combined equation of value, it is not unique. Thus the yield is undefined.

Solution 9

1. The equation of value is:

$$f(i) := 100\,(1+i)^4 - 20\,(1+i)^3 + 100\,(1+i)^2 - 20\,(1+i) - 170 = 0.$$

Denote the solution to the equation of value by $i^\star$. After trial and error, $f(0.01) = -4.736$ and $f(0.02) = +0.659$. Hence by linear interpolation between these two points,

$$i^\star \approx 0.01 + 0.01\,\frac{4.736}{4.736 + 0.659} = 0.01878.$$

2. Denote the first return in 2012–2013 by $i_1^\star$ and the second return in 2014–2015 by $i_2^\star$.

   For 2012–2013, the equation of value is: $f_1(i) := 100\,(1+i)^2 - 20\,(1+i) - 100 = 0$.

   As this is a quadratic equation, we solve it exactly to find that the only positive root is $1 + i_1^\star = 1.10499$.

   Do not linearly interpolate when you can determine the exact answer.

   For 2014–2015, the equation of value is: $f_2(i) := 200\,(1+i)^2 - 20\,(1+i) - 170 = 0$. Again, we solve it exactly to find that the only positive root is $1 + i_2^\star = 0.97331$.

   The (annual) LIRR is denoted LIRR. Hence, as there are two years in 2012–2013 and two years in 2014–2015,

$$(1 + \text{LIRR})^4 = (1 + i_1^\star)^2\ (1 + i_2^\star)^2 = 1.10499^2 \times 0.97331^2 \quad \Rightarrow \quad \text{LIRR } = 0.03706.$$

3. We calculate the growth factors for each period:

$$\begin{aligned}(1 + \text{TWRR})^4 &= \frac{110}{100}\,\frac{100}{(110-20)}\,\frac{180}{(100+100)}\,\frac{170}{(180-20)} = 1.16875\\ \Rightarrow \qquad &\text{TWRR } = 0.03975.\end{aligned}$$

Solution 10

1. Work in time units of half a year. The equation of value is:

$$f(i) := 100\,(1+i)^8 + 70\,(1+i)^5 - 50\,(1+i)^4 - 30\,(1+i)^3 - 140 = 0.$$

   Denote the solution to the equation of value by $i^\star$. After trial and error, $f(0.045) = -4.4181$ and $f(0.05) = +1.5812$. Hence by linear interpolation between these two points,

$$i^\star \approx 0.045 + 0.005\,\frac{4.4181}{4.4181 + 1.581} = 0.04868.$$

   Annualising this half-yearly interest rate, we find that the MWRR is 0.09973 per annum.

2. Denote the first return in 2012 by $i_1^\star$, the second return in 2013–2014 by $i_2^\star$ and the third return in 2015 by $i_3^\star$.

   For 2012, $i_1^\star = 105/100 - 1 = 0.05$ per annum.

   For 2013–2014, in time units of half a year, the equation of value is:

$$f_2(j) := 105(1+j)^4 + 70\,(1+j)^3 - 50\,(1+j)^2 - 30\,(1+j) - 130 = 0.$$

   Denote the solution to the equation of value by $j^\star$. After trial and error, $f_2(0.06) = -2.049$ and $f_2(0.07) = +4.042$. Hence by linear interpolation between these two points,

$$j^\star \approx 0.06 + 0.01\,\frac{2.049}{2.049 + 4.042} = 0.06336.$$

   Annualising the figure, $i_2^\star = 0.1307$ per annum.

   For 2015, $i_3^\star = \dfrac{140}{130} - 1 = 0.07692$.

   Hence $(1 + \text{LIRR})^4 = 1.05 \times 1.1307^2 \times 1.07692 \qquad \Rightarrow \qquad \text{LIRR } = 0.09652.$

3. We find growth factors for each period:

$$\begin{aligned}(1+\text{TWRR})^4 &= \frac{115}{100}\frac{195}{(115+70)}\frac{155}{(195-50)}\frac{140}{(155-30)} \\ \Rightarrow \quad \text{TWRR} &= 0.09758.\end{aligned}$$

Note: You could use the information in part 2 to find the TWRR. However, since no cashflows occur on 1 January 2013 or 1 January 2015, the growth factors would just cancel out.

Solution 11

1. The values of the funds before and after the cash injections are:

	Manager A		Manager B	
Date	Before	After	Before	After
1 Jan 2007	120 000		100 000	
31 Dec 2007	130 000	140 000	140 000	150 000
31 Dec 2008	135 000	145 000	145 000	155 000
31 Dec 2009	180 000		150 000	

Thus, TWRR for Manager A is given by:

$$(1+i_A)^3 = \frac{130}{120}\times\frac{135}{140}\times\frac{180}{145} \qquad \Rightarrow i_A = 0.09050$$

and the TWRR for Manager B is given by:

$$(1+i_B)^3 = \frac{140}{100}\times\frac{145}{150}\times\frac{150}{155} \qquad \Rightarrow i_B = 0.09409$$

2. MWRR for Manager A is given by:

$$120\times(1+i)^3 + 10\times(1+i)^2 + 10\times(1+i) = 180.$$

Then, putting $i = 0.094$ gives the left-hand side of the above equation equals 180.03, which is close enough to 180.

3. Both funds increased by 50% over the three-year period and received the same cashflows at the same times.

Since the initial amount invested in Manager B's fund was lower, the cash inflows received represent a larger proportion of Manager B's fund. Since Manager B's returns were negative for both the 2nd and 3rd years, whereas Manager A had only a negative return in the 2nd year (which was roughly of the same magnitude as Manager B's 2nd year return) the money-weighted return earned by Manager B over the period will be lower.

4. The money-weighted rate of return is higher for Manager A, whilst the time-weighted return is higher for Manager B. When comparing the performance of investment managers, the time-weighted rate of return is generally better because it ignores the effects of cash inflows or outflows being made which are beyond the manager's control.

In this case, Manager A's best performance is in the final year, when her fund was at its largest, whilst Manager B's best performance was in the first year, where his fund was at its lowest.

Overall, it may be argued that Manager B has performed slightly better than Manager A since Manager B achieved the higher time-weighted return.

Solution 12

1. • Either, calculate directly from the rates of return:
   The TWRR is i such that

$$(1+i)^2 = 1.01 \times 1.02 \times 1.05 = 1.08171 \qquad \Rightarrow \qquad i = 4.005\%.$$

   • Or, calculate the fund values just before each time a cashflow occurs.
   The fund value on 30 June 2009 will be $1.5 \times 1.01 = 1.515$.
   The fund value on 31 December 2009 will be $(1.515 + 6) \times 1.02 = 7.6653$.
   The fund value on 31 December 2010 will be $(7.6653 + 4) \times 1.05 = 12.2486$.
   The TWRR is i such that

$$(1+i)^2 = \frac{1.515}{1.5} \times \frac{7.6653}{7.515} \times \frac{12.2486}{11.6653} = 1.0817 \qquad \Rightarrow \qquad i = 4.005\%.$$

2. The equation of value is

$$1.5(1+i)^2 + 6.0(1+i)^{1.5} + 4(1+i) - 12.2486$$

   Trial-and-error results in: $f(4\%) = -0.1026$, $f(4.5\%) = -0.0210$ and $f(5\%) = 0.0607$.
   Interpolate between $i = 4.5\%$ and $i = 5\%$ to find

$$i = 0.045 + (0.05 - 0.045) \times \frac{0.0210}{0.0210 + 0.0607} = 0.0463.$$